

# Digital transformation is about to transform our society

Technology is no longer a matter of choice. It has now integrated into every aspect of our living. In India, the Covid-19-induced lockdowns have really accelerated investments in digital initiatives. The impact of these investments and initiatives is now visible. India is now the world's second-fastest-digitising economy and becoming a global force to reckon with.



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## India is leading the Digital Transformation Wave

FY23 provides a lot of promise for the global society as it rebounds from the impact of the pandemic. On one hand, businesses are leveraging digital networks to improve customer experience and gain competitive advantages. On the other hand, governments are using the power of technology to bridge the economic and digital divide and build connected cities and villages. Data, technology, and governance are coming together to accelerate socio-economic change like never before. It's remarkable to see how the Indian government is actively encouraging local manufacturing and R&D. This will help businesses leapfrog to Industry 4.0. These are indeed exciting times, as this change is being led by our billion strong populace. Another primary driver is the volume of data consumed, which is remarkable for a low per capita income country like India. India's per capita Internet usage has increased from 1.2 GB per month in 2018 to 14 GB per month in June 2021 according to the Economic Survey of India. This 12X growth in usage along with a multi-fold increase in internet penetration has set the foundation for the next phase of digital transformation.

### Some examples:

- The digital payment market will reach \$1 trillion by 2026
  - The digital health and telemedicine market will be worth over \$800 million by FY24
  - The Indian ecommerce market is expected to grow to US\$ 188 billion by 2025
  - OTT platforms are expected to reach \$13-15 Bn by 2030
  - Indian Edtech sector will reach US\$ 4 billion by 2025
- These use cases are just scratching the surface! New technologies like the Metaverse, Cryptocurrency, AI decision making, and Remote robotics will shape the future of entertainment, education, healthcare and commerce.

## The network will transform digital

Just like 8-lane expressways improve physical connectivity, fibre based broadband will transform digital connectivity. In that sense, digital networks are absolutely fundamental for these use cases to come alive. This network transformation is based on 3 build cycles: 5G, FTTx, and Rural and Open networks.

**5G** - The need for ubiquitous, high-speed connectivity along with very low network response times or latency, is driving 5G buildouts. Almost all the above mentioned use cases are hinging on 5G connectivity. According to the GSA, 489 operators in 146 countries/regions are investing in 5G testing, licensing, planning, network deployment. In India, policy frameworks are being set up and the countdown to spectrum allocations has begun.

**Fibre to the home** - Newer use cases like work from home, online education, telemedicine, and surging video consumption all point towards one thing: seamless connectivity at the end-points is now non-negotiable. That's the reason that FTTx build outs are gaining traction across countries including India, which has been a heavy mobile broadband market uptill now. As the world continues to battle with the pandemic and endeavours to keep up with

rapid modernisation, FTTx has emerged as one of the most impactful ways of delivering this seamless connectivity to millions of homes and premises..

**Rural and Open Networks** - For full digital transformation, rural and open networks are a must. Ubiquitous coverage will need high speed rural internet and governments are committing to the same. If this is to be done fast and efficiently, open and agile architectures should be implemented at scale. Open RAN opens up more ways to innovate and be more agile with expansion, pricing and services, while making the digital networks future ready in terms of scalability and superior control and management. This phenomena is picking up traction, not only in India but world over. According to Dell'Oro Group, cumulative Open RAN revenue from 2020 to 2025 might reach \$15 billion, accounting for more than 10% of the overall RAN industry by 2025.

Network creators viz - telecom operators, cloud firms, governments, and large enterprises are spending billions of dollars to build out these 3 types of networks in order to meet this demand. These large expenditures are being leveraged to fuel the above mentioned technologies, all of which are currently seeing rapid deployment and commercialization.

## STL playing a crucial role in India's Digital Economy

It is evident by the use cases and network architecture that we have discussed, the future of the network is converged. STL is playing an important role in designing and building this new architecture. On the optical side, we are enabling backhaul fiberisation, on the wireless side, we are developing Open RAN and programmable FTTx solutions. We are also building methods for high speed fibre deployment and building a talent pool for supporting this massive network modernisation world over.

Humanity's core strength is our ability to work together. India is at the precipice of having *Internet for everyone*. Imagine an India where everyone is connected and working together to make our country a better place. This transformation is happening now.